Fiscal Year:	FY 2013	**Task Last Updated:**	FY 09/01/2014
PI Name:	Laughlin, Mitzi Ph.D.		
Project Title:	Modulation of Muscle Function by Lower Limb Loading during Space Flight		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mslaughl@Central.UH.EDU	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-743-0405
Organization Name:	University of Houston		
PI Address 1:	Health and Human Performance		
PI Address 2:	3855 Holman St., Garrison Rm. 105R		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204-6015	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2010 Crew Health NNJ10ZSA003N
Start Date:	02/01/2012	**End Date:**	07/31/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Loerch, Linda	**Contact Phone:**	
Contact Email:	linda.loerch-1@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	Mitzi Laughlin replaced Bill Paloski due to his assignment with the Human Research Program, NASA Johnson Space Center		
COI Name (Institution):	Clarke, Mark (University of Houston) Layne, Chuck (University of Houston)		
Grant/Contract No.:	NNX12AF04G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Ed. NOTE (3/11/2013): Continuation of NNX12AF04G (PI=Paloski), per PI Paloski's assignment to Human Research Program, NASA Johnson Space Center. Muscle size, structure, and coordination changes during spaceflight are likely triggered by absent or reduced loading of receptors in the soles of the feet and the muscles and joints of the lower limbs. To better understand the physiological bases of these adaptive responses, we propose to examine, in-flight, the relationships between lower limb loading and muscle activation, and control. The proposed experiments will study astronaut volunteers before flight and on multiple occasions during flight using the MARES device to provide loading to the feet and legs as well as device that can apply pressure to the soles of the feet. The effects on muscle activation and performance associated with foot pressure alone or foot pressure plus lower limb loading will be investigated. If successful this investigation will provide new insight into the role of lower extremity load receptors in modulating muscle function and new evidence regarding the potential for a passive loading countermeasure to attenuate the undesirable side effects of space flight on muscle function.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The rescoping phase activities of “Modulation of Muscle Function by Lower Limb Loading during Space Flight” was completed by redesigning the proposed flight experiment into two separate experiments: 1) a smaller flight experiment that integrate in a cooperative/synergistic manner with the ESA Sarcolab Experiment, and 2) a bedrest flight analog experiment into a countermeasure feasibility evaluation study. The Principal Investigator for the flight study was Dr. Mitzi Laughlin and for the bedrest analog was Dr. Mark Clarke. The revised experiment protocols were delivered to the NASA Human Research Program Element Scientist Peter Norsk and his deputy Roni Cromwell. Additionally, we completed data collection for two ground-based supporting studies that will fine-tune and establish the reliability of the dependent measures protocols proposed for the motor performance and spinal cord excitability testing. These ground-based studies were used to refine our data reduction and analysis techniques. Both of these studies are in the data analysis stage now and results and publications will be shared with NASA and published in peer-reviewed scientific journals.
Bibliography Type:	Description: (Last Updated: 12/10/2020)